例 某单位负反馈闭环控制系统，其开环传递函数为

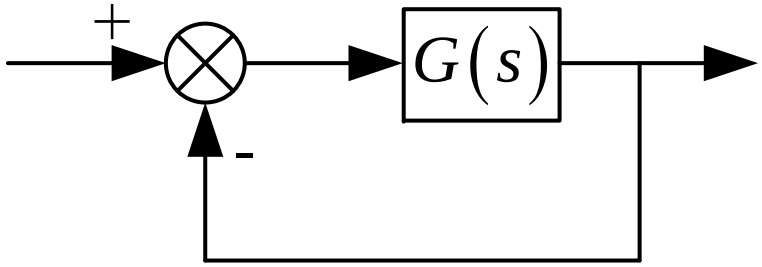
$$G(s)=\frac{200}{s(s+1)(s+20)}$$

(1) 画出其开环Nyquist图， 求Nyquist图与负实轴交点处的频率值$\omega_{-\pi}$；

(2) 判断闭环系统的稳定性，计算幅值裕量K_g（不必化为分贝值）；

(3) 画出开环对数幅频特性，根据折线图计算剪切频率ω_c；

(4) 根据(3)的结果，计算相位裕量。



$$G(j\omega)=\frac{200}{j\omega(j\omega+1)(j\omega+20)}$$

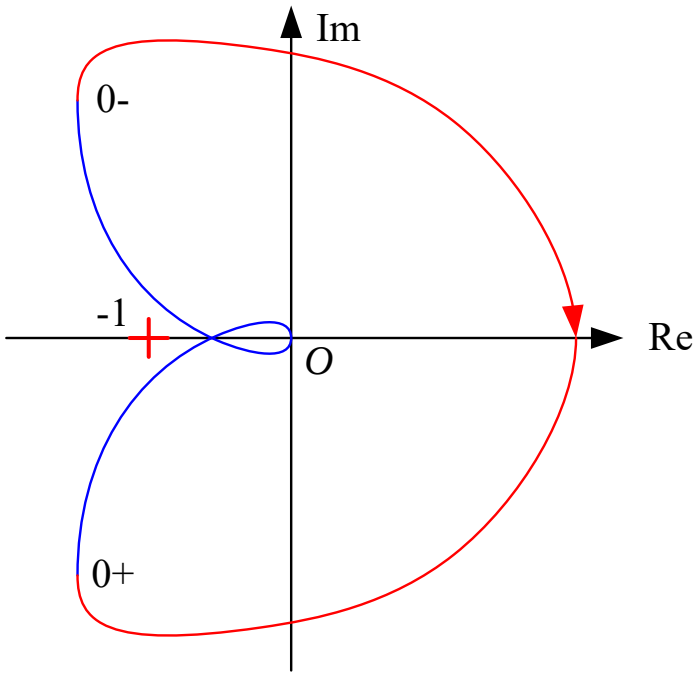
$$|G(j\omega)|=\frac{200}{\omega\sqrt{\omega^2+1}\sqrt{\omega^2+400}}$$

$$\angle G(j\omega)=-90^\circ-\arctan\omega-\arctan\frac{\omega}{20}$$

求Nyquist图与负实轴的交点:

$$\angle G(j\omega)=-90^\circ-\arctan\omega-\arctan\frac{\omega}{20}=-180^\circ$$

$$\Rightarrow \omega_{-\pi}=2\sqrt{5}\,\mathrm{rad/s}=4.47\,\mathrm{rad/s}\Rightarrow|G(j\omega_{-\pi})|=\frac{10}{21}$$

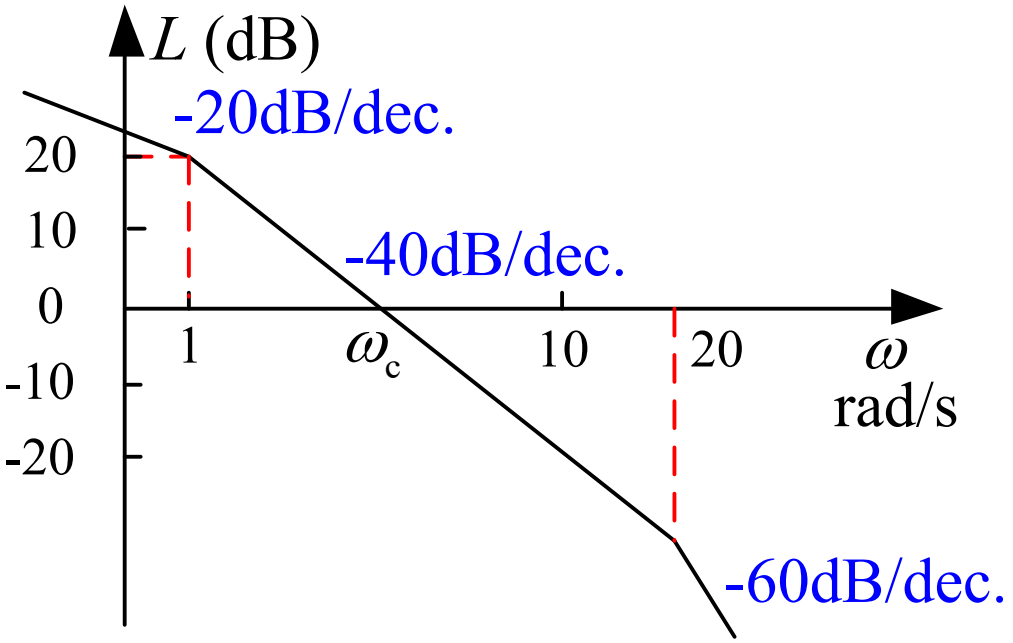


从Nyquist图可判断闭环系统稳定。

幅值裕量为 $K_g = \left| \dfrac{1}{G(j\omega_{-\pi})} \right| = 2.1$

$$G(s)=\frac{200}{s(s+1)(s+20)}$$

$$G(s)=\frac{10}{s(s+1)\left(\frac{1}{20}s+1\right)}$$



$$20\lg 10=40\lg\frac{\omega_c}{1}\Rightarrow \omega_c=\sqrt{10}\text{rad/s}\approx 3.16\text{rad/s}$$

相位裕量为:

$$\gamma=\angle G(j\omega_c)+180^\circ=90^\circ-\arctan(\omega_c)-\arctan(\frac{\omega_c}{20})=8.6^\circ$$